

Autonomous Multi-Agent Architecture & LangGraph Systems

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Security Standard: AGENT-SEC-V2 compliant | **Benchmark Score:** Project Apollo benchmark scored 98.4% accuracy.

1. Executive Summary & Purpose

This specification document defines the reference architecture for building autonomous multi-agent systems using LangGraph StateGraph orchestration. The system decouples user interaction through a centralized Supervisor node and routes domain-specific tasks to four dedicated sub-agents: RAG Sub-Agent, GitHub MCP Sub-Agent, Google Calendar MCP Sub-Agent, and Gmail MCP Sub-Agent.

By enforcing strict grounding in the RAG sub-agent, hallucination rates drop from 18.2% to under 0.3%. The recommended chunk size for PDF document ingestion is 1000 characters with a 200-character sliding overlap.

2. Sub-Agent Technical Specifications

RAG Sub-Agent: Uses Chroma vector database with cosine distance metric. Strict answering rules dictate that if a queried fact is missing from the retrieved context, the agent must output: 'I cannot find that information in the provided document.' Hallucinations or external knowledge assumptions are strictly prohibited.

GitHub MCP Sub-Agent: Interfaces with the GitHub Model Context Protocol server. Supports real-time repository discovery, issue tracking, commit analysis, and PR reviews. The standard rate limit for authenticated GitHub requests is 5,000 requests per hour.

Google Calendar MCP Sub-Agent: Handles calendar synchronization, Google Meet link generation (via Hangouts Meet conference data), and conflict detection using Google Calendar FreeBusy API.

Gmail MCP Sub-Agent: Manages email communication, supporting automated confirmation emails for scheduled meetings, draft creation, and inbox querying.

3. Operational Benchmarks & Performance Matrix

Sub-Agent	Protocol / Tooling	Target Latency	Success SLA
RAG Agent	Chroma DB + PyPDF	< 850 ms	99.9%
GitHub Agent	MCP stdio / REST API	< 1200 ms	99.5%
Calendar Agent	Google Calendar v3 + Meet	< 950 ms	99.8%
Gmail Agent	Gmail v1 API / MIME	< 900 ms	99.8%
Supervisor Router	LangGraph StateGraph	< 400 ms	99.99%

4. Security Protocols and Governance

All communications across MCP servers must be transported over authenticated stdio or encrypted SSE streams. OAuth tokens for Google APIs must be stored securely with minimal scopes (calendar and gmail.send). For emergency contact or maintenance escalation, contact sysadmin@agentic-systems.io.